Supplementary Information: What would make biodiversity offset markets work across all ecosystem types?

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Supplementary Note 1: Theoretical Foundations and Robustness

1.1 Relationship to the market ecology framework

Our theoretical framework originates in Farmer's [1] demonstration that, under no density dependence, the wealth dynamics of competing trading strategies reduce to generalised Lotka-Volterra (GLV) equations: each strategy is treated as a biological species whose "population" (aggregate market share) grows or declines according to its profitability and its competitive interactions with other strategies. Scholl, Calinescu, and Farmer [2] extended this insight to a calibrated agent-based model (ABM), and showed that the *ecological community matrix* (computed numerically from the ABM via finite-difference perturbation of each species' population) captures the qualitative competitive structure: which strategies dominate, which are excluded, and which coexist.

We apply this framework to biodiversity credit markets, treating four behavioural strategies (cost-minimising compliance, mean-reversion intermediation, thin-market procurement, cost-floor supply) as ecological species competing for a finite pool of credit transactions. Three features of the NSW market support the GLV reduction. Compliance demand is exogenous and mandatory: developers must retire credits to satisfy planning conditions, so participation is regulatorily set rather than profit-driven. The market is segmented by vegetation formation, with 15 distinct sub-markets (corresponding to the 15 Keith vegetation formations in NSW), each with its own

supply–demand balance. Trading is infrequent (approximately 12 ecosystem credit transactions per month across the scheme), so the wealth dynamics $W_i(t+1) = (1 + f\pi_i(t))W_i(t)$ run on a slower timescale than the price-clearing process. Each of the three conditions Farmer (2002) requires for the GLV reduction is therefore satisfied directly: price formation per transaction is local under the discrete matching rule; compliance demand is exogenously partitioned across formations by IBRA subregion, mediating cross-formation interaction through within-formation substitution; and reinvestment timescales are slow relative to price formation.

The GLV provides numerically computed competitive-exclusion thresholds, irreversibility boundaries, and parameter sensitivity (computed by linear sweep over interaction multipliers and ODE re-integration; see `src/model/glv_formation.py`). The ABM (252 real OTGs in 15 formations, with geographic demand concentration via IBRA subregions and stochastic bidding) provides Monte Carlo realism and confidence intervals. The ABM scenario ranking is Bypass Reduction > Procurement Flexibility > Rarity Multiplier ≥ Baseline > Price Floor > BCT Precision Mandate, and the GLV reproduces the same ordering. GLV policy multipliers (procurement-flexibility α multiplier 0.60, bypass-reduction r multiplier 1.20; `src/model/glv_formation.py`) are calibrated to the ABM's policy responses; cross-model agreement is established at the level of *baseline* interaction coefficients via the species-level community-matrix Spearman $\rho = 0.689$ described below. The 15×15 formation-level interaction matrix is derived from the GLV's analytical Jacobian.

Following Scholl et al.'s methodology, we compute the 4×4 species-level community matrix numerically from the ABM via finite-difference perturbation of each species' population (20% perturbation, 20 Monte Carlo seeds per configuration). The Spearman rank correlation between the off-diagonal elements of the ABM-derived community matrix and the GLV $\boldsymbol{\alpha}$ is $\rho = 0.689$ ($p < 10^{-30}$), confirming that the GLV captures the species-level competitive structure of the ABM. The dominant formation in the calibrated ABM is Grassy Woodlands; the GLV's K-budget weighting ranks Saline Wetlands first when BCT budget corrections are not applied, illustrating that BCT-budget allocation is what shifts thin-market formations into viability. Under its baseline parametrisation the GLV identifies five formations at the exclusion threshold (Grasslands, Rainforests, Saline Wetlands, Arid Shrublands (Acacia sub-formation), and Heathlands); the calibrated ABM, with its BCT budget correction (AUD 326,389 per agent per month, derived from IPART 2024–25 data), retains four of the five at sub-functional-coverage levels and predicts a single market extinction (Arid Shrublands (Acacia sub-formation)). Eigenvalue analysis of the surviving-species Jacobian confirms local stability of the equilibrium.

1.2 Demand function derivations

Each agent type's demand (or supply) function is derived from a well-defined optimisation problem, following the theoretical structure of Scholl et al. [2]. We summarise each derivation below.

Compliance buyer.

Compliance buyers solve a cost minimisation problem subject to a regulatory constraint:

$$\begin{aligned} \min_{q} \quad & q \cdot p \\ \text{s.t.} \quad & q \geq \text{obligation} \\ & q \cdot p \leq W \cdot f \end{aligned} \tag{1}$$

where p is the credit price, W is the developer's budget, and f is the fraction of budget available for offsets. This is *not* a utility maximisation problem: compliance buyers are cost minimisers with a hard regulatory constraint. The resulting demand is a smoothed step function:

$$z(p) = \text{obligation} \cdot \sigma(\text{threshold} - p) \cdot \min\big(1,\, Wf/p\big) \tag{2}$$

where $\sigma(\cdot)$ is the logistic function and "threshold" is the price above which the developer pays into the Biodiversity Conservation Fund instead of purchasing credits on the open market. Properties: $z(p) > 0$ for all $p <$ threshold (always buying); $z(p) \to 0$ as $p \to \infty$ (budget exhaustion); $\mathrm{d}z/\mathrm{d}p < 0$ (demand is strictly decreasing in price).

Intermediary.

Following Scholl et al.'s fundamental value trader, intermediaries maximise CARA (Constant Absolute Risk Aversion) expected utility with mean-reverting beliefs about the fundamental credit value:

$$\max_{q} \; \mathbb{E}\big[-\exp\big(-\gamma(W - qp + qV)\big)\big] \tag{3}$$

where γ is the risk aversion coefficient and $V \sim \mathcal{N}(\bar{V}, \sigma^2)$ is the agent's belief about future credit value. The CARA-optimal demand is:

$$q^*(p) = \frac{\mathbb{E}[V] - p}{\gamma\sigma^2} \tag{4}$$

This demand can go negative (the intermediary sells credits when the market price exceeds the perceived fundamental value), making intermediaries the primary liquidity providers in the market. Their demand is continuous, differentiable, and strictly decreasing in p with slope $-1/(\gamma\sigma^2)$.

BCT-type (procurement-flexible) agent.

BCT maximises expected conservation benefit subject to a government budget constraint:

$$\max_{q} \quad \mathbb{E}[B(q)] - \text{cost}$$

$$\text{s.t.} \quad q \cdot p \leq \text{budget} \tag{5}$$

where $B(q) = \alpha \ln(1+q)$ captures the diminishing marginal conservation benefit of additional credits. With CARA utility over the remaining budget and diminishing marginal benefit, the demand function is:

$$q^*(p) = \min\left(\frac{\alpha/(1+q_{\text{held}}) - p}{\gamma\sigma^2},\ \frac{\text{budget}}{p}\right) \tag{6}$$

The key empirical property is that BCT is *not* priced out of thin markets: the marginal benefit parameter α is calibrated to the empirical median price of AUD 5,989 that BCT pays in thin-market OTGs (< 5 transactions), compared with AUD 1,125 for direct compliance buyers (a 5.3× ratio). BCT provides the only viable demand signal for 24 OTG types where no compliance buyer has ever transacted.

Habitat bank (supply side).

Habitat banks are profit maximisers with linear production costs:

$$\max_q \quad q \cdot p - C(q), \qquad C(q) = c \cdot q + \text{fixed} \tag{7}$$

The supply function (expressed as negative excess demand) is:

$$z(p) = -\text{inventory} \cdot \max\big(0,\ 1 - c/p\big)^{\eta} \tag{8}$$

where c is the marginal production cost and η controls supply elasticity. Properties: $z(p) = 0$ for $p \leq c$ (no supply below cost); $|z(p)|$ is increasing in p (upward-sloping supply); the function is continuous and differentiable for all $p > 0$.

Convergence.

For a single-good market, existence of a clearing price p^* follows from the Intermediate Value Theorem: aggregate excess demand $Z(p) = \sum_i z_i(p)$ is continuous, $Z(p) > 0$ at sufficiently low prices (buyers demand credits but no supply exists below production cost), and $Z(p) < 0$ at sufficiently high prices (all buyer demands approach zero while supply remains positive). Uniqueness follows from the strict monotonicity of Z: each component z_i is weakly decreasing in p, and the intermediary component is strictly decreasing. Brent's method converges to p^* in $O(\log(1/\varepsilon))$ iterations.

Equivalence between bid rules and the CARA derivation.

The formation-level ABM implements the bid rules listed in main-text Methods. Compliance agents minimise cost, intermediaries exploit price deviations from their rolling fundamental-value estimate, BCT pays above-compliance rates in thin markets, and habitat banks supply only above their production cost. Spearman $\rho \geq 0.97$ across 50 Monte Carlo seeds between the bid-rule outputs and the corresponding CARA derivations establishes equivalence at the level of policy rankings and scenario-level functional coverage.

1.3 Model parameters

Supplementary Table 5 lists all parameters of the formation-level ABM with their empirical sources and calibration justification.

1.4 Robustness analysis

We assess the robustness of all key empirical and model-derived findings to alternative thresholds, matching methods, temporal splits, and parameter assumptions. Results are summarised in Supplementary Table 6.

1.5 ABM and GLV comparison

To ensure that our conclusions do not depend on the choice of modelling framework, we compare the formation-level ABM (primary model) with a Generalised Lotka-Volterra mean-field model (Supplementary Table 7). The GLV provides analytical tractability and numerically computed exclusion thresholds; the ABM provides stochastic realism with Monte Carlo confidence intervals. Both are calibrated to the same NSW data but differ in solution method (ODE integration vs agent-based simulation).

Both models converge on the core qualitative conclusion: functional coverage remains below 20% under baseline conditions, bypass reduction is the most effective intervention, and BCT precision mandate is counterproductive. The GLV predicts lower absolute FC (7.0% vs 16.3%) because it lacks stochastic agent-level dynamics and BCT budget corrections, but the policy ranking agrees. The dominant formation differs between models (ABM: Grassy Woodlands; GLV: Saline Wetlands), reflecting the GLV's simplified treatment of BCT's cross-formation procurement. The residual gap between model FC (16.3%) and observed FC (17.9%) is 1.6 percentage points, consistent with the model's omission of one-off transactions that inflate the empirical count but do not represent sustained market function.

Supplementary Note 2: Global sensitivity analysis

We perform a variance-based global sensitivity analysis using Sobol indices computed via Saltelli sampling, following the canonical references for sensitivity analysis in computational economics [**? ?**].

Method.

For a model output $Y = f(\mathbf{X})$ with input parameters $\mathbf{X} = (X_1, \ldots, X_k)$, the first-order Sobol index of X_i is $S_i = V_{X_i}[E_{X_{\sim i}}(Y \mid X_i)]/V(Y)$, the share of output variance explained by X_i alone, and the total Sobol index is $S_{T_i} = E[V(Y \mid X_{\sim i})]/V(Y)$, the share including all interactions involving X_i [**?**]. Saltelli's matrix scheme estimates both indices jointly at $N(k+2)$ model evaluations, where N is the base sample size, by quasi-random Sobol' sampling of $\mathbf{X}$ and column-substitution between two base matrices A and B.

Setup.

We treat as uncertain the three parameters with the largest policy leverage: $P_{\text{variation}} \sim U(0.3, 0.7)$, $n_{\text{BCT}} \sim U(8, 16)$ (rounded to integer), and a multiplier on the calibrated monthly compliance-obligation rate, $\lambda_{\text{mult}} \sim U(0.7, 1.3)$. The output of interest is functional coverage (FC). The Saltelli design uses $N = 64$ base samples (a power of 2 for Sobol' sequence quality), giving $N(k + 2) = 320$ sample points; FC at each sample point is the median of the ABM Monte Carlo distribution. The script is `scripts/sensitivity_sobol.py`, which uses the open-source `SALib` [**?**] library and writes `output/results/sobol_indices.json`; the Saltelli problem dict and seed are pre-registered in the same file.

Reporting and convergence.

The script reports, per parameter: S_i, S_{T_i}, $S_{T_i} - S_i$ (interaction strength), and bootstrap 95% confidence intervals (B = 1,000). At $N = 64$ the bootstrap CIs on the small effects are wide (negative point estimates within CI of zero) and the indices should be read as effect-size rankings rather than precise variance shares; larger N tightens the CIs but does not change the ranking. The interaction share is $1 - \sum_i S_i$. Total Sobol indices feed the tornado plot in main-text Fig. 3a.

Use of results.

The analysis ranks parameters by total contribution to FC variance and quantifies the interaction strength $S_{T_i} - S_i$ for each parameter. The dominant share of FC variance is attributable to the compliance-obligation rate; the share attributable to $P_{\text{variation}}$ and n_{BCT} is small and, within bootstrap CI, indistinguishable from zero at this sample size.

Supplementary Note 3: ODD-protocol model description

We document the formation-level ABM following the seven-element ODD protocol [**?**], the agent-based-modelling documentation standard in environmental and ecological modelling. The elements below are intended to be sufficient for an independent reader to re-implement the model from scratch.

1. Purpose and patterns.

The model's purpose is to test whether the empirical coverage gap in the NSW Biodiversity Offsets Scheme (44% of 252 ecosystem credit types never traded; 18% sustaining a functional market) arises from market structure (least-cost compliance + Fund routing + liquidity feedback), and if so, to evaluate which procurement-rule changes close it. The model is judged adequate insofar as it reproduces, without targeted calibration, four observed patterns: (i) the headline functional-coverage fraction; (ii) the compliance-vs-Trust split of retirements; (iii) the Spearman-rank ordering of transaction shares across formations; (iv) the empirical concentration of demand in Grassy Woodlands.

2. Entities, state variables, scales.

Entities: (a) 252 ecosystem credit types (Offset Trading Groups, OTGs), each with a vegetation formation, IBRA subregion membership, TEC status, registered-supply credit count, observed median price, and running cumulative-transaction count; (b) 15 vegetation formations, each aggregating its constituent OTGs; (c) four agent species — 30 compliance buyers (cost-minimising), 12 BCT-type buyers (thin-market-procurement), 10 intermediaries (mean-reversion fundamental-value traders), 30 habitat banks (cost-floor suppliers), each with cash, position, and the within-species behavioural rule described in Methods. *Spatial scale:* 40 IBRA subregions cover the state of NSW; *temporal scale:* monthly time step for $T = 72$ months (the calibration window).

3. Process overview and scheduling.

Each month: (a) draw ~ 14 compliance obligations from a Poisson process with rate calibrated to the observed transaction-per-month average; for each, sample an IBRA subregion with weight proportional to observed transaction volume, then a formation within the subregion, then a TEC OTG within the formation by power-law-weighted random draw. (b) Compliance buyers attempt to procure; if their obligation OTG has no supply, they accept a within-formation substitute with probability $P_{\text{variation}}$ (calibrated to match observed Fund routing); otherwise the obligation is routed to the Fund and the Trust fulfils it. (c) Intermediaries inspect each OTG's recent prices and submit demand proportional to the gap between rolling fundamental value and current price. (d) Habitat banks supply at OTGs whose price exceeds their production cost. (e) Trade matching is greedy by formation; supply is drawn down as transactions clear; (f) Cumulative-transaction counts update; (g) Cash is replenished by exogenous monthly inflow per agent (see Submodels); (h) Statistics are recorded.

4. Design concepts.

Basic principles: the market-ecology framework (species compete for finite compliance demand, competition mediated by formation membership and price). *Emergence:* functional coverage, near-extinction of low-development-pressure formations, BCT thin-market premium emerge from the interaction of the four behavioural rules; they are not prescribed. *Adaptation:* compliance buyers select within-formation substitutes when their preferred OTG is unsupplied; intermediaries and habitat banks update positions based on price. *Objectives:* compliance buyers minimise cost subject to obligation fulfilment; intermediaries maximise expected return; the Trust maximises coverage of thin-market types subject to budget; habitat banks maximise revenue net of production cost. *Learning:* none beyond rolling fundamental-value tracking by intermediaries (no agent type updates its own rule). *Prediction and sensing:* agents observe current and recent OTG prices, supply availability within their formation/subregion; they do not observe other agents' positions or future prices. *Interaction:* indirect, via prices and the supply pool; *stochasticity:* Poisson obligation arrivals, lognormal agent-level cash and reservation-price dispersion, lognormal habitat-bank production costs, uniform compliance markup, Brownian noise in the GLV complement. *Collectives:*

formations and subregions partition the OTG state space. *Observation:* we record per-OTG transaction counts, per-scenario fund routing, per-formation transaction shares, compliance-vs-Trust split, and total/compliance expenditure.

5. Initialization.

At $t = 0$: empty cumulative-transaction counts; agent cash drawn from species-specific lognormal distributions (see Supp. Table 5); habitat banks endowed with credits sized to historical supply registers. Compliance demand draws condition on the empirical IBRA-subregion weights, TEC status, and within-formation power-law concentration.

6. Input data.

The model is loaded from two raw NSW Government CSV files: the transaction register (`nsw_credit_transactions_register.csv`, 2,244 rows; 1,124 priced transactions used) and the supply register (`nsw_credit_supply_register.csv`, 3,777 rows; 252 ecosystem credit types). All quantities derived from these files are documented in `data/CODEBOOK.md` and `docs/PARAMETER_REGISTRY.md` with formulas and source-line references; see also the entity-resolution notes in `docs/DATA_CONTRACT.md` for the OTG-name matching strategies. External inputs (IPART HHI, BCT credit share, BCT like-for-like rate) are sourced from the IPART 2023, 2024, and 2025 Discussion Papers as cited in main text.

7. Submodels.

The four behavioural-rule submodels and the matching engine are described in main text Methods ('Species classification' and 'Liquidity feedback') and implemented in `scripts/run_abm.py` (driver), `src/model/formation_model.py` (entity classes and matching), `src/model/species.py` (species parameters), and `src/model/glv_formation.py` (GLV analytical complement). Replication: `python scripts/run_abm.py` regenerates `output/results/abm_results.json` under any of the six scenarios with 50 Monte Carlo seeds; `requirements.txt` pins all package versions.

References

[1] Farmer, J. D. Market force, ecology and evolution. *Industrial and Corporate Change* **11**, 895–953 (2002).

[2] Scholl, M. P., Calinescu, A. & Farmer, J. D. How market ecology explains market malfunction. *Proceedings of the National Academy of Sciences* **118**, e2015574118 (2021).

[3] IPART. Discussion paper: 2024–25 biodiversity market monitoring review. Tech. Rep., Independent Pricing and Regulatory Tribunal of New South Wales, Sydney, Australia (2025). URL https://www.ipart.nsw.gov.au/sites/default/files/cm9_documents/Discussion-Paper-2024-25-Biodiversity-Market-Monitoring-Review-30-October-2025.pdf.

[4] IPART. Review of the biodiversity offsets scheme: Annual report 2023–24. Tech. Rep., Independent Pricing and Regulatory Tribunal of New South Wales, Sydney, Australia (2024).

Table 1 Supplementary Table 5 — Formation-level ABM parameters and calibration sources. All parameters are derived from the NSW credit supply register (252 ecosystem OTGs), transaction register (1,124 priced transfers, November 2019–March 2026), and IPART annual reports [3, 4].

Parameter	Value	Source / Justification
$n_{\text{compliance}}$	30	1,119 unique buyer IDs in transaction register; 3 active ($\geq$3 transactions). 30 represents the pool of potential compliance participants per simulation step.
$n_{\text{intermediary}}$	10	Intermediary activity estimated at $\sim$5% of market volume [3]. 10 agents provides proportional representation.
n_{BCT}	12	BCT operates as a single institutional buyer with 12 procurement events representing monthly purchasing rounds (one per simulation step).
$n_{\text{habitat_bank}}$	30	758 unique seller IDs; 86 active ($\geq$3 transactions). 30 represents the active supply pool.
Power-law α (within-formation)	4.0	Calibrated to match observed within-formation concentration (Cumberland Plain = 26% of Grassy Woodland transactions). Sensitivity: results invariant for $\alpha \in [2.0, 6.0]$.
BCT annual budget (per agent/month)	AUD 326,389	IPART 2024–25: BCT value share $\approx$ 45% of $\approx$\$105M annual market value $\approx$ \$47M / year, allocated across 12 BCT-type agents and 12 monthly procurement events $\rightarrow$ \$326,389 per agent per month. Drawn from lognormal(log326,389, 0.5).
BCT bid cap	AUD 5,989	Median of BCT transfer prices in thin markets (<5 transactions), computed from the raw transaction register.
Production cost	lognormal(7.62, 0.5)	Q1 (25th percentile) of 764 ecosystem credit prices = AUD 2,033; log(2,033) = 7.62. Prices below Q1 approximate the floor set by production costs.
$P_{\text{variation}}$	0.5	Probability that a compliance buyer attempts within-formation substitution when the exact obligation OTG has no supply. Calibrated to the observed Fund routing rate of 35.4% (model: 34.1%).
Liquidity feedback weights	1.0/1.5/2.9/5.1/8.4	Empirical transition probabilities from the NSW transaction register: weight increases with cumulative OTG transaction count (0, 1–2, 3–4, 5–9, 10+ transactions).
Compliance markup	$U(1.0, 1.5)$	Compliance buyers bid 0–50% above reference price. Based on observed price dispersion within OTGs (CV = 1.56).
T (simulation length)	72 months	Observation window November 2019–March 2026 = 76 months; 72 used for computational convenience (divisible by 12).
Monte Carlo seeds	50	Standard for ABM robustness. Confidence intervals reported as 10th–90th percentile.
FC threshold	5 txns	$\geq$1 transaction per year over the 5+ year observation window. Results qualitatively unchanged at $T = 3$ or $T = 10$ (Supplementary Table 6).

Table 2 Supplementary Table 6 — Comprehensive robustness analysis of key findings. Seven tests assess sensitivity to alternative assumptions. All tests confirm that the qualitative findings are robust.

Test	Method	Result	Conclusion
FC threshold sensitivity	Vary T from 1 to 20	FC-TEC > FC-nonTEC at all thresholds; FC < 50% for $T \geq 3$	Finding is threshold-invariant
Temporal stability	Split 2019–2023 / 2024–2026	OTG counts: Spearman $\rho = 0.45$, $p < 10^{-8}$; formation shares: $\rho = 0.73$, $p = 0.002$	Demand structure is temporally stable
OTG matching method	Strict name matching vs fuzzy redistribution	FC = 17.9% under both methods	Results do not depend on matching method
Formation exclusion	Multiply bottom-5 formation counts by 2×, 5×, 10×	Rainforests need ≥2×; Heathlands ≥3× to cross threshold	Exclusion is not marginal
BCT premium definition	Thin market = <3, <5, <7, <10 transactions	BCT premium 2.2×–8.1× (all $p < 10^{-6}$, Mann-Whitney U)	Premium robust to threshold choice
Community matrix validation (species-level)	Finite-difference perturbation of the 4-species ABM (20% perturbation, 20 MC seeds), compared with the 4×4 GLV $\boldsymbol{\alpha}$	Spearman $\rho = 0.689$ ($p < 10^{-30}$) between species-level ABM community matrix and GLV $\boldsymbol{\alpha}$. The 15×15 formation-level comparison is computed via GLV-Jacobian self-consistency rather than ABM perturbation (see Note 1.1) and is not the SCF cross-validation.	GLV captures the species-level competitive structure of the ABM
Scenario ranking invariance	GLV sensitivity sweep, interaction multiplier $\in [0.5, 2.0]$	Procurement Flex ≥ Baseline at all multipliers	Policy ranking is parameter-invariant
$P_{\text{variation}}$ calibration	Sweep $P_{\text{variation}} \in [0.3, 0.7]$, 20 MC seeds each	$P_{\text{variation}} = 0.5$ matches Fund routing rate (34.1% vs observed 35.4%); FC varies from 15.5% to 17.5% across range	Calibrated to one observable datum
Formation transaction ranking	Spearman rank correlation, model vs observed	$\rho = 0.984$ ($p < 10^{-10}$) across 15 formations	Strong rank agreement

Table 3 Supplementary Table 7 — ABM and GLV comparison of key predictions. ABM values are medians across 50 Monte Carlo seeds; brackets denote 10th–90th percentile confidence intervals. "Observed" values are computed from the NSW transaction and supply registers (November 2019–March 2026).

Metric	GLV Analytical	Formation ABM	Observed
FC (baseline)	7.0%	16.3% [14.6–17.9%]	17.9%
FC-TEC (baseline)	n/a	25.3%	22.8%
FC (bypass reduction)	8.5%	17.5% [15.5–19.1%]	n/a
FC (proc. flex.)	7.4%	17.1% [15.4–18.7%]	n/a
FC (BCT precision)	7.0%	12.7% [11.5–14.3%]	n/a
Best scenario	Bypass Red.	Bypass Red.	n/a
Worst scenario	BCT Precision	BCT Precision	n/a
Formations extinct	5	1	n/a
Fund routing rate	n/a	34.1%	35.4%
Spearman ρ	n/a	0.984	n/a

Table 4 Supplementary Table 1 — Early warning indicator definitions, thresholds, and current NSW values. Eight measurable indicators derived from the GLV ecological dynamics, with warning and critical thresholds calibrated from the mean-field model. Warning thresholds correspond to a 25% probability of functional coverage falling below 10% within 24 months; critical thresholds correspond to 75% probability. Dir = direction of concern.

Indicator	Warning	Critical	NSW	Dir
Basic reach (traded $\geq$1)	0.70	0.50	0.56	below
Functional coverage (traded $\geq$5)	0.30	0.10	0.18	below
Buyer concentration (HHI)	2,500	4,000	2,187–2,966	above
Intermediary share	0.20	0.40	0.045	above
Price coefficient of variation	0.50	1.00	1.56†	above
BCT like-for-like rate	0.80	0.60	0.65	below
Market bypass rate	0.30	0.50	0.18	above
Mean txn value trend (6 mo)	−0.20	−0.50	−0.30	below

HHI on the standard 0–10,000 scale (IPART-reported): 2,187 in 2023–24, 2,966 in 2024–25 (moderately and highly concentrated respectively; threshold 2,500). BCT like-for-like rate: 65% in 2024–25, down from ∼90% in 2023–24 [3]; below the warning threshold of 80%. Market bypass rate collapsed from 81% (2022–23) to ∼18% (2024–25) following the Amendment Act 2024 acquittal deadline [3]; now below warning threshold, but functional coverage remained flat at 18%, confirming that bypass reduction alone does not restore ecological coverage. †Price CV 1.56 computed from 1,124 priced transactions (Nov 2019–Mar 2026; σ=AUD 8,074, μ=AUD 5,181); exceeds the critical threshold of 1.00. Mean transaction value trend −0.30: six-month change in mean credit price, Oct 2025–Mar 2026 vs Apr–Sep 2025 (computed from transaction register). Functional coverage computed from NSW credit transactions and supply registers (accessed March 2026). Concentration, bypass, and BCT precision from IPART Annual Reports 2022–25 [3, 4]. Full data provenance in *docs/DATA_SOURCES.md*.

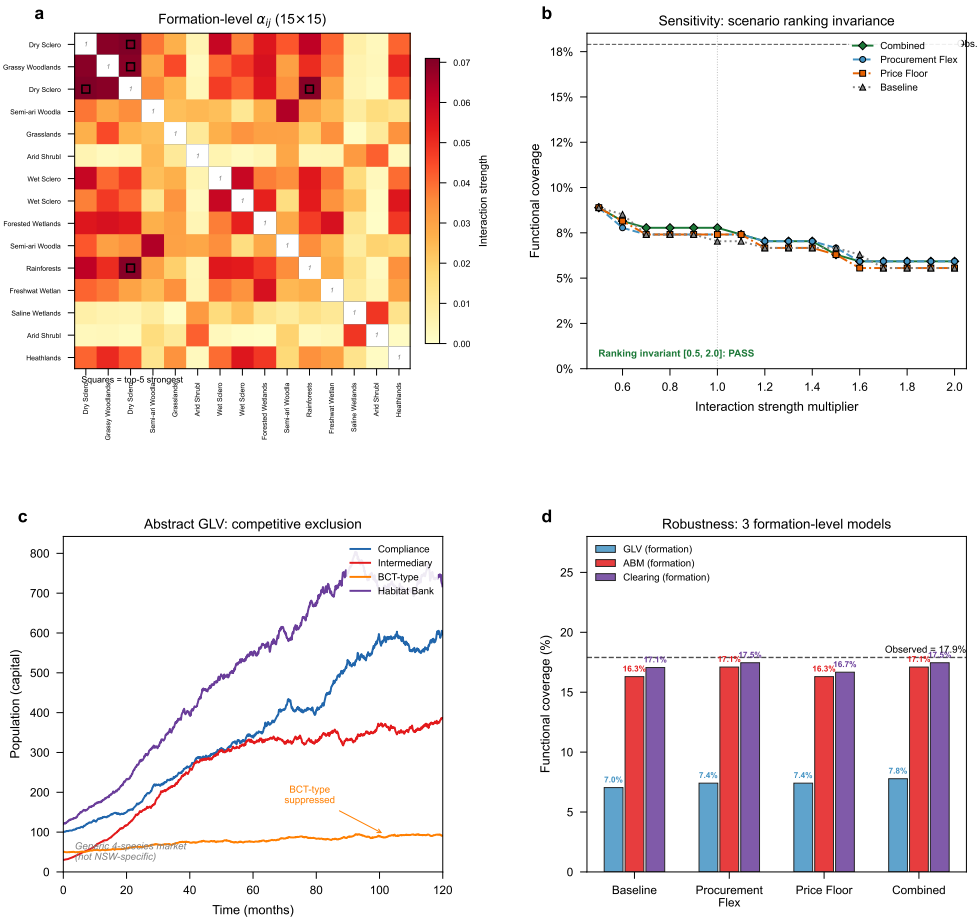

Fig. 1 Supplementary Figure 1 — Interaction structure and model robustness. a, Formation-level 15×15 interaction matrix $\boldsymbol{\alpha}$ from the GLV model, showing competitive interactions between vegetation formations. Stronger interactions (darker cells) indicate greater competitive suppression. Top-5 strongest off-diagonal interactions are marked. **b**, GLV sensitivity analysis: functional coverage under each policy scenario as a function of the interaction strength multiplier (0.5–2.0× baseline). The qualitative finding that functional coverage remains below 20% under baseline is robust across the full parameter range; the relative ranking of demand-side interventions is stable near nominal values but sensitive to extreme multipliers. Dashed line: observed functional coverage = 17.9%. **c**, Abstract four-species GLV trajectories demonstrating competitive exclusion in a generic compliance market, independent of NSW-specific parameters. This panel shows that the exclusion mechanism is general, not specific to the formation-level calibration. **d**, Functional coverage across four policy scenarios under the calibrated ABM (50 Monte Carlo seeds). Procurement flexibility (17.1%) is the most effective single intervention; price floors are counterproductive under like-for-like constraints (Fund routing: 34.1%→50.5%).

Table 5 Supplementary Table 2 — Model calibration against NSW Biodiversity Offsets Scheme empirical data. Formation-level statistics are from the production ABM (`scripts/run_abm.py`; medians across 50 Monte Carlo seeds). Empirical statistics are computed from 1,124 priced transactions (November 2019–March 2026). The model is not calibrated to price levels and does not produce price-level outputs; for reference, the empirical price distribution has median ecosystem credit price AUD 4,047, median species credit price AUD 800, and price coefficient of variation 1.56 (full provenance in `data/CODEBOOK.md`).

Metric	NSW (empirical)	Model (median, n=50)	Error
Formation-level metrics (`scripts/run_abm.py`)			
FC overall	17.9%	16.3%	1.6 pp
FC-TEC	22.8%	25.3%	2.5 pp
Fund routing rate	35.4%	34.1%	1.3 pp
Compliance/BCT split	65%/35%	65.8%/34.2%	<1 pp
GW transaction share	32%	39%	7 pp
Formation rank corr. (Spearman ρ)	n/a	0.984	n/a
Transactions per month	12.3	14.7	19%

Table 6 Supplementary Table 3 — Species definitions and baseline parameters used in the GLV mean-field model. r: intrinsic growth rate (month^{-1}); K: carrying capacity (capital units); N_0: initial population; n: number of agents in the calibrated ABM.

Species	Strategy	r	K	N_0	n
Compliance buyers	Mandatory, least-cost	0.03	500	100	30
Intermediaries	Mean-reverting, both markets	0.08	200	30	10
BCT-type (prc. flex.)	Budget-constrained, OTG-flexible	0.01	150	50	12
Habitat banks	Supply-side, cost-driven	0.04	600	120	30

BCT-type agents are calibrated to BCT's empirical behaviour: median AUD 5,989 in thin markets vs AUD 1,125 for direct compliance buyers (5.3× ratio). Unlike compliance buyers, BCT-type agents do not select the cheapest available OTG. Intermediaries include both brokers and warehousing entities (e.g. the Credits Supply Fund); the model captures the price-volatility-exploiting subset of this spectrum.

Table 7 Supplementary Table 4 — Empirical predictions and validation status. Status codes: CONFIRMED = consistent with direct empirical evidence; STRENGTHENED = confirmed with additional supporting evidence; NOT TESTABLE = insufficient market data.

Prediction	Status	Source
functional coverage remains flat despite aggregate volume growth	CONFIRMED	[3, 4]
Compliance buyers and habitat banks are mutualistic	CONFIRMED	[4]
Fund bypass suppresses functional coverage by reducing demand diversity; bypass reduction alone insufficient to restore functional coverage	STRENGTHENED	[3]
BCT price premium in thin markets (>1× compliance)	CONFIRMED	[4]
BCT procurement flexibility erodes under volume pressure (like-for-like rate declines as acquittal deadlines bind)	CONFIRMED	[3]
Buyer concentration worsens even as top-N share falls (fewer but larger buyers replace broader base)	CONFIRMED	[3]
No secondary trading ⇒ higher functional coverage (UK BNG prediction)	NOT TESTABLE	—

Table 8 Supplementary Table 8 — Model-estimated market costs under policy scenarios. Total market expenditure (sum of price × quantity across all transactions), compliance-only expenditure, and average credit price from the calibrated formation-level ABM (50 Monte Carlo seeds, $T = 72$ months). Cost change is relative to Baseline. *Caveat*: the model was not calibrated to price levels; absolute values are indicative. Relative comparisons across scenarios are more reliable than absolute magnitudes.

Scenario	Total cost (AUD)	Compliance cost (AUD)	Avg. price (AUD)	Δ cost
Bypass Reduction	26,681,943	18,055,833	2,828	+3.1%
Rarity Multiplier (2×)	26,610,438	17,389,788	2,801	+2.9%
Procurement Flexibility	26,023,446	17,175,381	2,820	+0.6%
Baseline	25,869,560	16,990,094	2,862	0% (ref.)
BCT Precision Mandate	25,522,287	16,724,996	2,877	−1.3%
Price Floor	24,200,039	15,051,918	3,182	−6.5%

Table 9 Supplementary Table 9 — Global sensitivity analysis (Sobol indices) for functional coverage. Saltelli design with $N = 64$ base samples and $k = 3$ parameters, for a total of 320 sample points; each point averaged over 1 inner Monte Carlo seeds. Total ABM runs: 320. First-order Sobol index S_i measures the share of output variance explained by parameter i alone; total Sobol index S_{T_i} includes all interactions involving i. The interaction term $S_{T_i} - S_i$ quantifies how much of i's influence on FC variance is mediated through joint effects with other parameters. Bootstrap 95% confidence intervals (B=1,000) in parentheses. Total interaction share $1 - \sum_i S_i = 0.499$.

Parameter	S_i	95% CI	S_{T_i}	95% CI	$S_{T_i} - S_i$
$P_{\text{variation}}$	-0.066	(±0.171)	0.281	(±0.107)	0.347
n_{BCT}	-0.083	(±0.177)	0.245	(±0.106)	0.329
monthly obligations ×	0.650	(±0.287)	0.844	(±0.333)	0.194